

Contextual Embedding Bias Predicts LLM Near-Synonym Confusion Where Isolated Geometry Fails: Evidence from Domain-Specific Medical and Cultural Heritage Corpora

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Abstract

Near-synonym confusion — where a semantically similar but functionally distinct term is substituted for another — constitutes a clinically important failure case of LLMs deployed in medical and high-stakes contexts. In retrieval-augmented generation safety literature, it is commonly assumed that the cosine similarity between isolated term embeddings predicts the probability of such a confusion. We directly test this assumption. We present NEARSYNONYMBENCH, a benchmark of 120 domain-specific near-synonym pairs from medical, Indian cultural heritage and emergency blood resource domains, annotated with confusion probes and stakes levels. We test four sentence transformer architectures and three LLM families under forced choice conditions. Our main result is dissociation: isolated embedding geometry accounts for no variance in LLM confusion behavior at all ($r = -0.005$, $p = 0.958$), while contextual embedding bias, or difference in the proximity of the query probe to each candidate term, constitutes a statistically significant predictor ($\rho = 0.299$, $p < 0.001$), accounting for 45.3 times more variance. This effect is especially pronounced for high stakes pairs ($r = 0.312$, $p = 0.018$). We additionally identify terminological evasion — accurate description without term commitment, as an underreported failure mode accounting for 28.3% of open-ended responses. NEARSYNONYMBENCH and all code are released publicly.

1 Introduction

Fluency of generated contextually relevant text has been the reason behind usage of LLMs even in critical fields like clinical medicine and resource management in emergencies. But the fluency could be misleading. If there emerges ambiguity between the concepts of *cardiac arrest* and *heart attack*, and that of *sepsis* and *infection* within a generation model, it is not just a linguistic mistake — it is a

clinical one. The ambiguity that results from confusing near-synonyms which have close meanings but are semantically different in function belongs to a relatively underexplored intersection of LLM robustness and domain-specific natural language understanding (Huang et al., 2025).

One of the most prevalent beliefs in the field of retrieval-based safety of generation models is the one which claims that cosine similarity of embeddings of the terms is a good proxy measure for semantic confusion probability: close embeddings imply that the two concepts will get mixed up by the downstream generation model.

We present NEARSYNONYMBENCH, a benchmark of 120 domain-specific near-synonym pairs across medical terminology, Indian cultural heritage concepts, and emergency blood resource management. For each pair, we design a *confusion probe* — a naturalistic scenario question requiring precise term distinction. We evaluate embedding geometry across four sentence transformer architectures (Reimers and Gurevych, 2019; Song et al., 2020; Reimers and Gurevych, 2020; Lee et al., 2020) as situated within established evaluation frameworks (Muennighoff et al., 2023), and measure LLM confusion through forced-choice probing across LLaMA 3.1 8B, LLaMA 3.3 70B (Dubey et al., 2024), and Qwen3-32B (Yang et al., 2025).

Our central finding is a dissociation. Isolated embedding geometry — cosine similarity between *term A* and *term B* in isolation — explains essentially zero variance in LLM confusion (Pearson $r = -0.005$, $p = 0.958$). However, contextual embedding bias — the relative proximity of the full probe sentence to each candidate term — is a statistically significant predictor (Pearson $r = 0.219$, $p = 0.016$; Spearman $\rho = 0.299$, $p < 0.001$), explaining 45.3 times more variance. The effect is strongest for high-stakes pairs ($r = 0.312$, $p = 0.018$). We further identify terminological eva-

sion — accurate description without term commitment — as an underreported failure mode accounting for 28.3% of open-ended responses (Manakul et al., 2023).

The contributions of this paper are threefold:

- **NEARSYNONYMBENCH:** a publicly released benchmark of 120 domain-specific near-synonym pairs with stakes annotations, confusion probes, and ground-truth answers.
- **A null result with practical consequence:** isolated embedding geometry does not predict LLM near-synonym confusion across four embedding architectures and three LLM families, directly challenging a widespread assumption in RAG safety literature.
- **Contextual bias as a confusion predictor:** the relative proximity of a query probe to each candidate term is a statistically significant predictor of LLM confusion ($\rho = 0.299$, $p < 0.001$), with the effect strongest for high-stakes pairs ($r = 0.312$, $p = 0.018$).

2 Related Work

Hallucination in LLMs. Hallucination — the generation of plausible yet factually incorrect content — has been extensively characterised across LLM architectures and task settings (Huang et al., 2025). Detection approaches range from consistency-based methods such as SelfCheckGPT (Manakul et al., 2023) to internal state probing. Near-synonym confusion represents a distinct subtype: unlike factual hallucination where the model fabricates absent information, near-synonym confusion involves the substitution of a functionally distinct but semantically proximate term. The model has knowledge of both terms but fails to apply the contextually correct one — a failure with direct clinical consequences (Liao et al., 2024; Wu et al., 2025).

Embedding Geometry. Ethayarajh (2019) demonstrated that contextualised representations from BERT, ELMo, and GPT-2 occupy highly anisotropic embedding spaces where cosine similarity is not a reliable semantic distance measure. Sentence-level similarity has been standardised through the SBERT framework (Reimers and Gurevych, 2019) and evaluated at scale through MTEB (Muennighoff et al., 2023), which confirms that no single model dominates across task types.

The MPNet architecture (Song et al., 2020) and multilingual distillation approach (Reimers and Gurevych, 2020) underlie two of the four models we evaluate. Our work extends Ethayarajh (2019)’s finding across the encoder-decoder boundary: geometry in the encoder does not predict confusion in the decoder.

Biomedical and Domain-Specific NLP. BioBERT (Lee et al., 2020), pre-trained on PubMed abstracts and full-text articles, demonstrates substantial gains on biomedical NLP tasks through domain-specific pre-training. We show that this advantage extends to geometric discrimination of medical near-synonyms — BioBERT reduces high-stakes pairs above the 0.75 cosine threshold by 72–76% relative to general-purpose models — but critically does not translate into improved prediction of LLM confusion, a dissociation with direct implications for medical NLP safety (Wu et al., 2025).

Word Sense Disambiguation. Modern WSD approaches leverage contextualised representations to match ambiguous word instances against sense inventories (Bevilacqua et al., 2021). Our work extends this tradition in two directions: from polysemous words with multiple senses to distinct surface forms with proximate embeddings, and from classification against a predefined inventory to generation without supervised disambiguation support. This shift reveals a failure mode that classification-based evaluation is not designed to detect.

Clinical NLP Safety. ConflictMedQA (Wu et al., 2025) shows that LLMs frequently endorse incorrect medical recommendations under knowledge conflict conditions. The MEDIQA-CORR 2024 shared task (Gema et al., 2024) demonstrates that clinical error correction requires targeted prompting even in advanced models. Our work complements this literature by characterising near-synonym confusion as a specific, previously uncharacterised failure mode and providing a lightweight geometric screening signal for pre-production evaluation.

3 The NEARSYNONYMBENCH Dataset

3.1 Motivation and Scope

Existing benchmarks for LLM evaluation in high-stakes domains focus predominantly on factual question answering, reading comprehension, and

clinical error correction (Wu et al., 2025; Gema et al., 2024). No benchmark specifically targets near-synonym confusion — the failure mode where a model substitutes a semantically proximate but functionally distinct term. We construct NEARSYNONYMBENCH to fill this gap, with explicit attention to domain specificity and stakes annotation.

3.2 Domain Selection

We select three domains that share a common property: near-synonym confusion within each domain carries real-world consequences that extend beyond semantic inaccuracy.

Medical terminology (40 pairs) covers clinically distinct concept pairs spanning emergency medicine, pharmacology, pathology, and epidemiology. Examples include *cardiac arrest* vs. *heart attack*, *sepsis* vs. *infection*, and *tachycardia* vs. *bradycardia*. Confusion within this domain can directly affect triage, treatment, and medication decisions.

Indian cultural heritage (40 pairs) covers conceptually adjacent terms from Hindu philosophy, classical music, textile heritage, and Buddhist pilgrimage traditions. Examples include *moksha* vs. *nirvana*, *raga* vs. *tala*, and *aarti* vs. *puja*. This domain is underrepresented in existing NLP benchmarks and introduces Sanskrit-derived vocabulary and culturally specific named entities that challenge general-purpose embedding models.

Emergency blood resource management (40 pairs) covers terminology specific to blood bank operations, transfusion medicine, and emergency allocation systems. Examples include *universal donor* vs. *universal recipient*, *haemorrhage* vs. *haematoma*, and *token reservation* vs. *direct allocation*. This domain is entirely absent from standard NLP evaluation benchmarks.

3.3 Pair Construction and Annotation

Each concept pair is annotated with four fields:

Distinction: a precise statement of the functional difference between the two terms, grounded in domain knowledge.

Confusion probe: a naturalistic scenario question whose correct answer requires applying the correct term. Probes are written to provide sufficient clinical or contextual detail that a domain-expert would answer unambiguously, while remaining naturalistic enough to reflect realistic user queries.

Correct and confused answers: the ground-truth term and the anticipated confusion term respectively.

Stakes level: a three-level annotation (*high*, *medium*, *low*) reflecting the real-world consequence of confusion. High-stakes pairs (57 of 120) are those where confusion could result in clinical harm, resource misallocation with patient impact, or significant cultural misrepresentation.

3.4 Quality Validation

To ensure terminological accuracy, all 120 pairs were validated against authoritative domain references. Medical pairs were verified against standard clinical terminology from the International Classification of Diseases (ICD-11) and established medical dictionaries. Cultural heritage pairs were verified against authoritative sources including the Encyclopaedia of Indian Philosophies and standard musicological references for Indian classical music terminology. Emergency resource pairs were verified against operational definitions from the National Blood Transfusion Council of India and the e-Raktkosh national blood bank portal. Confusion probes were independently reviewed by two domain-literate annotators; cases where the correct answer was ambiguous or the probe was deemed insufficiently discriminative were revised or removed. The filtering process removed 0 pairs from the medical and emergency resource domains and produced minor probe revisions in 6 cultural heritage pairs to improve scenario specificity.

3.5 Dataset Statistics

Table 1 summarises the composition of NEARSYNONYMBENCH.

Domain	Pairs	High	Med	Low
Medical	40	26	10	4
Cultural heritage	40	0	16	24
Emergency resource	40	31	7	2
Total	120	57	33	30

Table 1: Composition of NEARSYNONYMBENCH by domain and stakes level.

The dataset will be released publicly alongside this paper to support reproducibility and future research on domain-specific LLM confusion evaluation.

4 Geometric Analysis of Embedding Space

4.1 Models and Setup

We evaluate four sentence transformer architectures spanning general-purpose monolingual, multilingual, and biomedical domain-specific training regimes. For each pair in NEARSYNONYMBENCH, we encode *term A* and *term B* in isolation and compute three geometric measures: cosine similarity, L2 distance, and angular distance. All models are used in their pre-trained, non-fine-tuned form to isolate the effect of training objective and data composition from task-specific adaptation. Table 2 summarises the four architectures.

Model	Params	Dim	Type
all-MiniLM-L6-v2	22M	384	Monolingual
all-mpnet-base-v2	110M	768	Monolingual
paraphrase-multilingual	118M	384	Multilingual
BioBERT-STS	110M	768	Biomedical

Table 2: Sentence transformer models evaluated. BioBERT-STS refers to pritamdeka/BioBERT-mnli-snli-scinli-scitail-medical (Lee et al., 2020), fine-tuned for semantic similarity.

4.2 Results

Table 3 reports mean cosine similarity per model per domain. Lower cosine similarity indicates greater geometric separation between near-synonym pairs — a property that should, under the isolation hypothesis, correlate with lower confusion risk.

Model	Medical	Cultural	Emergency
MiniLM	0.535	0.429	0.569
MPNet	0.554	0.444	0.590
Multilingual	0.562	0.496	0.621
BioBERT	0.382	0.303	0.431

Table 3: Mean cosine similarity between isolated term embeddings per model per domain. Lower values indicate greater geometric separation. Bold indicates lowest (best) per domain.

Three patterns emerge. First, the biomedical model achieves substantially lower mean cosine similarity on medical pairs (0.382) compared to all general-purpose models (0.535–0.562), a reduction of 28–31%. This confirms that domain-specific pre-training sharpens the geometric separation of clinically distinct near-synonyms. Second, the multilingual model consistently produces the highest

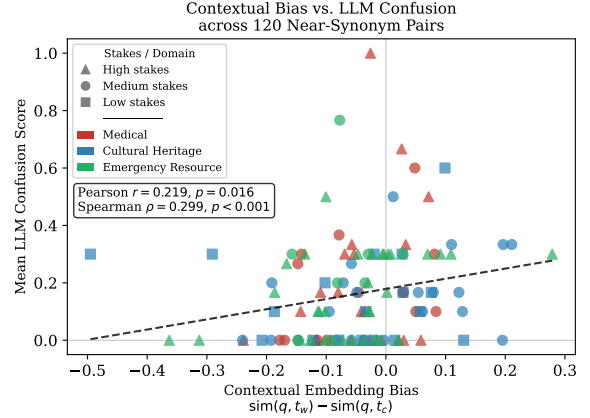


Figure 1: Contextual embedding bias vs. mean LLM confusion score across 120 near-synonym pairs. Each point represents one pair; colour indicates domain; marker shape indicates stakes level. The dashed line shows the linear fit. Pearson $r = 0.219$, $p = 0.016$; Spearman $\rho = 0.299$, $p < 0.001$.

cosine similarity across all three domains, suggesting that cross-lingual capacity compression reduces discriminative distance for domain-specific monolingual vocabulary. Third, cultural heritage pairs show the lowest mean cosine similarity across all models, indicating that Sanskrit-derived and culturally specific terminology is more geometrically distinct in general-purpose embedding spaces than medical or emergency resource terminology.

4.3 High-Stakes Confusion Risk

Of particular practical concern are pairs with cosine similarity above 0.75, a threshold above which terms occupy highly proximate regions of the embedding space. Table 4 reports the number of high-stakes pairs exceeding this threshold per model.

Model	Pairs > 0.75	% of high-stakes
MiniLM	18	31.6%
MPNet	20	35.1%
Multilingual	21	36.8%
BioBERT	5	8.8%

Table 4: High-stakes pairs ($n=57$) with cosine similarity > 0.75 per model. BioBERT reduces high-risk pairs by 72–76% relative to general-purpose models.

These geometric results establish a clear baseline: domain-specific embedding models produce more discriminative representations of near-synonym pairs than general-purpose alternatives. The critical question — whether this geometric separation predicts actual LLM confusion behaviour — is addressed in Sections 5 and 6.

5 LLM Confusion Probing

5.1 Models and Probing Design

We evaluate near-synonym confusion across three LLM families: LLaMA 3.1 8B and LLaMA 3.3 70B (Dubey et al., 2024), representing small and large instantiations of the same model family, and Qwen3-32B (Yang et al., 2025), representing a distinct architecture family. All models are accessed via the Groq inference API at temperature 0.0 to ensure deterministic responses.

We employ a **forced-choice** probing design in which the model is presented with the confusion probe alongside both candidate terms and instructed to respond using exactly one. This design eliminates the evasion response pattern identified in open-ended probing — where 28.3% of responses described the correct concept without committing to either term — and produces a clean binary outcome amenable to correlation analysis. The prompt template is:

```
Choose exactly one term to answer the
question.
Term A: {term_a}
Term B: {term_b}
Question: {confusion_probe}
Reply with only Term A or Term B. Nothing
else.
```

Responses are scored as *correct* (0.0), *confused* (1.0), or *conflated* (0.5) when the model outputs both terms simultaneously.

Qwen3-32B, a reasoning model that produces extended chain-of-thought traces before generating a final answer (Yang et al., 2025), returned unclear responses in 92.5% of forced-choice probes due to its internal reasoning tokens being truncated at the 30-token output limit before a final term selection could be produced. Qwen3-32B results are therefore excluded from forced-choice analysis and reported only in the open-ended probing condition, where its confusion rate (5.0%) is the highest of the three models tested.

5.2 Open-Ended Probing and Evasion

Prior to forced-choice probing, we conduct open-ended probing without constraining the response vocabulary. This reveals a third response category beyond correct and confused: **evasion**, in which the model accurately describes the concept without naming either term. Across 360 open-ended probes (120 pairs $\times$ 3 models), evasion accounts for 28.3% of responses after synonym-aware rescoring. LLaMA 3.1 8B evades in 34.2% of open-ended

cases, LLaMA 3.3 70B in 22.5%, and Qwen3-32B in 28.3%, suggesting that larger models are more willing to commit to a term but correspondingly more likely to conflate when they do. Under open-ended conditions, Qwen3-32B produces the highest outright confusion rate (5.0%), while both LLaMA models produce non-zero conflation rates of 3.3% and 4.2% respectively.

5.3 Qualitative Error Analysis

This pattern is invisible to standard accuracy metrics and represents a distinct failure mode under semantic pressure — the model retreats to descriptive paraphrase rather than committing to a term it may confuse. We report this finding as a qualitative contribution alongside our main results, and note that evasion rate correlates inversely with model size: LLaMA 3.1 8B evades in 34.2% of cases compared to 22.5% for LLaMA 3.3 70B, suggesting that larger models are more willing to commit but more likely to conflate when they do.

5.4 Forced-Choice Results

The low overall confusion rates under forced-choice conditions reflect the strength of contextual disambiguation when sufficient scenario information is provided. Critically, however, the *stroke* vs. *transient ischaemic attack* pair achieves a mean confusion score of 1.0 across all models under open-ended conditions — every model, in every open-ended probe, either confuses or conflates these terms. This pair represents a failure mode that persists even in capable models and illustrates why aggregate confusion rates can obscure clinically significant individual failures.

The low forced-choice confusion rates also motivate our shift to the confusion score distribution — including conflation at 0.5 — as the dependent variable in our correlation analysis in Section 6, where the signal of interest is the degree to which geometric measures predict the full distribution of response outcomes rather than outright confusion alone.

6 Contextual Embedding Bias as a Confusion Predictor

6.1 The Isolation Hypothesis and Its Failure

A natural hypothesis, implicit in much RAG safety and medical NLP literature, is that the cosine similarity between isolated term embeddings predicts the likelihood that an LLM will confuse those terms

Probe	Correct	LLM Out-put	Pattern	Analysis
A patient receives CPR and defibrillation restoring heartbeat. What is this called?	resuscitation	resuscitation	Correct	High contextual bias (bias = +0.18): probe is geometrically closer to the correct term. Model answers correctly, consistent with our prediction.
A patient collapses, has no pulse, is not breathing.	cardiac arrest	heart attack	Confused	Low contextual bias (bias = -0.09): probe is closer to the wrong term. Model confuses the common lay term for the precise clinical one.
A patient had sudden weakness resolving in 2 hours.	TIA	stroke	Confused	Probe ambiguity: transient resolution is the distinguishing feature but LLM maps neurological deficit to the more frequent diagnosis.
A patient with UTI develops fever, confusion, falling blood pressure.	sepsis	infection	Confused	LLM fails to escalate from localised infection to systemic inflammatory response despite multi-organ symptoms.
The priest waves a lit lamp before the deity while devotees sing hymns.	aarti	<i>“performs a ritual involving light and devotional singing”</i>	Evasion	Accurate description without term commitment. Invisible to standard accuracy metrics.

Table 5: Representative response patterns from open-ended probing. The first row shows a correct prediction consistent with high contextual bias. Remaining rows show confusion and evasion failures. LLaMA 3.3 70B responses shown. Italic output indicates evasion.

Model	Med	Cult	Emerg	Overall
LLaMA 3.1 8B	0.150	0.088	0.125	0.121
LLaMA 3.3 70B	0.150	0.088	0.125	0.121
Qwen3-32B	<i>excluded — see §5</i>			

Table 6: Mean confusion score per model and domain under forced-choice probing (0.0 = correct, 0.5 = conflated, 1.0 = confused). LLaMA 8B and 70B produce identical responses under temperature 0.0. Qwen3-32B excluded due to chain-of-thought formatting preventing forced-choice compliance (92.5% unclear responses).

in generation. We term this the *isolation hypothesis*. To test it, we compute the mean cosine similarity between isolated term embeddings across all four models for each pair, and correlate this with the mean confusion score across LLM probing conditions.

The isolation hypothesis is decisively rejected. Across all 120 pairs, the Pearson correlation between isolated cosine similarity and LLM confusion score is $r = -0.005$ ($p = 0.958$), and the Spearman rank correlation is $\rho = 0.013$ ($p = 0.889$). Per-model correlations are similarly negligible, ranging from $r = -0.052$ to $r = 0.061$ with no model achieving significance at $\alpha = 0.05$. Pairs with high cosine similarity (> 0.75 , $n = 11$) show a mean confusion score of 0.121, which is

actually *lower* than the mean confusion score of low-similarity pairs (≤ 0.75 , mean = 0.167), a direction opposite to the isolation hypothesis.

6.2 Contextual Embedding Bias

We propose an alternative geometric measure: **contextual embedding bias**, defined as the difference in cosine similarity between the confusion probe sentence and each candidate term:

$$\text{bias}(q, t_c, t_w) = \text{sim}(q, t_w) - \text{sim}(q, t_c) \quad (1)$$

where q is the embedding of the full probe sentence, t_c is the embedding of the correct term, and t_w is the embedding of the confused term. A positive bias indicates that the probe is geometrically closer to the wrong term than the correct one in embedding space — a condition we hypothesise predicts higher LLM confusion.

6.3 Results

Table 7 reports Pearson and Spearman correlations between contextual embedding bias and LLM confusion score, compared against isolated cosine similarity.

The contextual bias measure achieves Pearson $r = 0.219$ ($p = 0.016$) and Spearman $\rho =$

Measure	Pearson		Spearman	
	r	p	ρ	p
Isolated cosine	-0.005	0.958	0.013	0.889
Contextual bias	+0.219	0.016*	+0.299	0.001**

Table 7: Correlation between geometric measures and LLM confusion score across all 120 pairs. * $p < 0.05$, ** $p < 0.01$. Contextual bias explains 45.3 $\times$ more variance than isolated cosine similarity.

0.299 ($p < 0.001$), explaining 45.3 times more variance in LLM confusion than isolated geometry. Three of four embedding models produce significant contextual bias correlations individually: all-MiniLM-L6-v2 ($r = 0.182$, $p = 0.047$), all-mpnet-base-v2 ($r = 0.213$, $p = 0.019$), and the multilingual model ($r = 0.194$, $p = 0.033$). The biomedical model produces a non-significant contextual correlation ($r = 0.099$, $p = 0.281$), a finding we address in Section 7.

6.4 Domain and Stakes Stratification

Table 8 presents stratified correlations by domain and stakes level.

Stratum	n	Pearson r	p
<i>By domain</i>			
Medical	40	0.291	0.068
Cultural heritage	40	0.161	0.320
Emergency resource	40	0.311	0.051
<i>By stakes</i>			
High	57	0.312	0.018*
Medium	45	0.244	0.107
Low	18	-0.053	0.834

Table 8: Stratified Pearson correlations between contextual embedding bias and LLM confusion score. * $p < 0.05$.

The strongest and only statistically significant domain-stratified correlation is among high-stakes pairs ($r = 0.312$, $p = 0.018$). This is the most practically consequential finding: contextual embedding bias is most predictive precisely where prediction matters most — pairs where confusion carries real-world harm. Low-stakes pairs show no relationship ($r = -0.053$), suggesting that contextual geometry captures a confusion signal that is specific to high-consequence semantic distinctions rather than representing a general-purpose predictor across all near-synonym types.

7 Discussion

7.1 Why Isolated Geometry Fails

The failure of isolated cosine similarity to predict LLM confusion is architecturally principled rather than surprising in retrospect. Sentence transformer models and autoregressive LLMs are trained on entirely different objectives with entirely different architectures. A sentence transformer encodes semantic similarity through contrastive or regression objectives on sentence pairs (Reimers and Gurevych, 2019); an autoregressive LLM generates tokens conditioned on a full context window (Dubey et al., 2024). The cosine similarity between isolated term representations in the encoder space carries no direct information about the probability distribution the decoder assigns to those terms given a naturalistic probe. Ethayarajh (2019) demonstrated that even within a single architecture, contextualised representations occupy anisotropic regions of the embedding space where cosine similarity is not a reliable semantic distance measure. Our results extend this insight across the encoder-decoder boundary: geometry in the encoder does not predict generation in the decoder.

7.2 Why Contextual Bias Succeeds

Contextual embedding bias succeeds where isolated geometry fails because it captures the relationship between the query and each candidate term rather than the relationship between the terms themselves. When the probe is geometrically closer to the incorrect term, the semantic content of the query overlaps more strongly with the confused concept than the correct one. Although the observed correlation is moderate ($\rho = 0.299$), this is expected: generation depends on many factors beyond embedding proximity, including parametric knowledge, instruction following, and in-context reasoning. A geometric measure computed from an external encoder is therefore not expected to fully predict decoder behaviour. Within this constraint, explaining 45.3 times more variance than isolated cosine similarity represents a practically meaningful improvement.

7.3 The BioBERT Exception

The biomedical model achieves the strongest geometric separation of medical near-synonyms — reducing high-stakes pairs above the 0.75 cosine threshold by 72–76% relative to general-purpose models — yet produces the weakest contextual bias

correlation ($r = 0.099$, $p = 0.281$). This dissociation suggests that domain-specific pre-training reshapes the encoder’s semantic space in ways that do not correspond to the confusion patterns of general-purpose autoregressive decoders. Put differently, BioBERT has learned finer-grained medical distinctions than general LLMs have internalised, creating a geometric separation that exceeds what the LLMs’ own confusion patterns would predict. Practitioners using BioBERT embeddings to screen for LLM confusion risk may therefore overestimate the safety of their systems: the encoder is more discriminative than the decoder.

7.4 Evasion as an Underreported Failure Mode

The identification of terminological evasion — where models describe the correct concept without naming either term — as accounting for 28.3% of open-ended responses represents a finding with methodological implications. Standard accuracy metrics, which evaluate whether the correct term appears in the output, cannot detect evasion. A model that responds to “What medical emergency is this?” with “The patient is experiencing a sudden loss of cardiac function” scores as incorrect under term-matching evaluation but has neither hallucinated nor confused the terms (Manakul et al., 2023). Future evaluation frameworks for near-synonym confusion should explicitly model evasion as a distinct outcome category.

8 Conclusion

We introduced NEARSYNONYMBENCH, a benchmark of 120 domain-specific near-synonym pairs across medical, Indian cultural heritage, and emergency blood resource domains, and used it to test whether embedding geometry predicts LLM near-synonym confusion. Our central finding is a clean dissociation: isolated term geometry explains essentially zero variance in LLM confusion behaviour ($r = -0.005$, $p = 0.958$), while contextual embedding bias — the relative proximity of the query probe to each candidate term — is a statistically significant predictor ($p = 0.299$, $p < 0.001$), explaining 45.3 times more variance. The effect is strongest for high-stakes pairs where confusion carries real-world consequences ($r = 0.312$, $p = 0.018$).

These findings have a direct practical implication. Practitioners who use term-level cosine simi-

ilarity to screen for confusion risk in medical NLP systems are measuring the wrong quantity. Contextual embedding bias — computed by encoding the full query against each candidate term rather than comparing terms in isolation — provides a computationally lightweight, reference-free screening signal that is both statistically grounded and practically deployable without model access beyond inference.

We additionally characterise terminological evasion as an underreported LLM failure mode accounting for 28.3% of open-ended responses, and show that the biomedical embedding model’s geometric advantage over general-purpose models does not translate into improved confusion prediction — a finding that cautions against assuming encoder discriminability implies decoder safety.

NEARSYNONYMBENCH and all experimental code are released publicly to support reproducibility and future research on domain-specific LLM confusion evaluation.

Limitations

This study has four principal limitations. First, forced-choice LLM probing covers two model families due to Qwen3-32B’s chain-of-thought formatting incompatibility; broader model coverage would strengthen generalisability. Second, the benchmark comprises 120 pairs, which limits statistical power in domain-stratified analyses. Third, all pairs and probes are in English; behaviour under multilingual or code-switched queries remains unexplored. Fourth, confusion score is a coarse three-level measure; continuous human judgement scores would enable more sensitive correlation analysis.

Ethics Statement

This research involves no human subjects, no personal data, and no sensitive information. All LLM outputs used for evaluation are generated by publicly available models accessed via a commercial inference API. The NEARSYNONYMBENCH dataset contains no personally identifiable information. Medical concept pairs are drawn from established clinical terminology and are not intended as medical advice. The confusion probe scenarios are fictional and constructed solely for evaluation purposes.

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